

XAI Agent Analysis — CREDIT DOMAIN

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RESPONSIBLE AI SCORE		
95 / 100		A ■ EXCELLENT
Explainability	Fairness	Compliance
30/30	40/40	25/30

Model Type	Domain	Accuracy	Test Samples
XGBoost	CREDIT	79.5%	200

EXECUTIVE SUMMARY

This Responsible AI Audit was conducted using the XAI Agent system, which combines SHAP explainability, LIME verification, fairness analysis using Fairlearn and IBM AIF360, and GDPR/EU AI Act compliance checking.

The model achieved an overall **Responsible AI Score of 95/100** with a status of **READY FOR DEPLOYMENT** . The EU AI Act classifies this system as **HIGH RISK**, requiring conformity assessment and human oversight before deployment.

Key findings include a fairness score of **100/100** indicating gender-based bias patterns in predictions, and a compliance score of **82/100** with most GDPR articles satisfied due to the explainability layer.

KEY FINDINGS

Finding	Status	Priority
SHAP + LIME explanations available	COMPLIANT	—
Gender bias detected (sex_A93, sex_A94)	FAIL	CRITICAL
GDPR Article 22 (Right to Explanation)	COMPLIANT	—
GDPR Article 25 (Privacy by Design)	AT RISK	HIGH
EU AI Act Risk Classification	HIGH RISK	CRITICAL
DPIA completed and documented	COMPLIANT	—

BIAS ANALYSIS

Overall Fairness Score: **100/100**

Protected Attribute	DPD	DPD Status	EOD	EOD Status
sex_A92	0.0	PASS 	0.0	PASS 
sex_A93	0.0	PASS 	0.0	PASS 
sex_A94	0.0	PASS 	0.0	PASS 

PROXY FEATURES DETECTED

Feature	Protected Attribute	Correlation	Risk Level
none	all	0.0	LOW RISK 

COMPLIANCE ANALYSIS

Compliance Score: **82/100** | EU AI Act Risk: **HIGH**

Regulation	Severity	Status	Finding
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REMEDIATION ROADMAP

Prioritized actions to improve the Responsible AI Score:

#	Priority	Action	Regulation
1	CRITICAL	Implement bias mitigation for sex_A93 and sex_A94	FAIRNESS
2	HIGH	Monitor proxy features: age, dependents correlation	FAIRNESS
3	MEDIUM	Add human review workflow for HIGH RISK decisions	EU_AI_ACT

This report was generated automatically by Rithvik's XAI Agent System. All findings should be reviewed by a qualified AI ethics professional before deployment.